Output Form (OF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: *INJONGO* | ***Context:*** *Urban Ethiopian Utility Chatbot Users — Amharic-Language Telegram/SMS*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

LOWER-priority dimension with strong alignment. Both benchmark and deployment use structured text output (intent label + slot extraction) with standard text representation. Intent detection is scored by accuracy and slot-filling by F1 [Q84], and the structured `$$`-separated extraction format [Q146] is a reasonable text representation that maps to programmatic parsing. Models are instructed to return `$$` only when no entities are found [Q145], handling the empty-output case explicitly. Five seeds for fine-tuning [Q63] and five prompt templates for LLM eval [Q73] provide robustness. The output modality is text-only, matching deployment exactly. Minor concerns: the `$$` separator is benchmark-specific syntax that requires post-processing in production, and the text-string output form can complicate confidence-score extraction useful for chatbot dialogue management. F1 metric is appropriate for span extraction.

ID	Evidence
Q63	"All results are averaged over five seeds." (p.5)
Q73	"The evaluations of LLMs use 5 different prompting templates and a temperature of 0.5." (p.6)
Q84	"Evaluation is based on accuracy and F1-score for ID and SF tasks. Average computed on five templates, and on only African languages." (p.7)
Q145	"Please ensure that the entities match the listed types and that unstated entities should not be included in the response if no entities are found, return `$$` only." (p.19)
Q146	"Sentence: John went to Paris and paid 100 dollars at an Awater restaurant. Output: PERSONAL_NAME John $$ CITY_OR_PROVINCE Paris $$ MONEY 100 $$ RESTAURANT_NAME Awater" (p.19)

Strengths

- Intent accuracy and slot F1 are standard, well-understood metrics [Q84]
- Five-seed averaging for fine-tuning and five-template averaging for LLMs reduce evaluation variance [Q63, Q73]
- Structured output format with explicit empty-case handling [Q145, Q146]
- Text-only modality matches Telegram/SMS deployment exactly
- Both fine-tuning and prompting paradigms evaluated [Q61, Q65], allowing the user to choose deployment-appropriate models

ID	Evidence
Q61	"We evaluate three categories of models: (1) encoder-only models such as XLM-RoBERTa Large (Conneau et al., 2019), AfroXLMR (Alabi et al., 2022), AfroXLMR-76L (Adelani et al., 2023a), AfriBERTa V2 (Oladipo, 2024), (2) encoder-decoder models such as mT5-Large (Xue et al., 2020), AfriTeVa V2 Large (Oladipo et al., 2023), and (3) a multilingual variant of LLM2Vec model (BehnamGhader et al., 2024) i.e. NLLB-LLM2Vec (Schmidt et al., 2024) that stack NLLB-encoder (NLLB Team et al., 2022) model with LLaMa 3.1 8B (Grattafiori et al., 2024) to develop a multilingual sentence transformer model." (p.5)
Q63	"All results are averaged over five seeds." (p.5)
Q65	"First, we conduct zero-shot prompting using the following widely used LLMs for evaluation: GPT4o, Gemini 1.5 Pro (Reid et al., 2024), Gemma 2 9B/27B Instruct (Team et al., 2024), Llama 3.1 8B/3.3 70B Instruct (Grattafiori et al., 2024), and Aya-101 (Üstün et al., 2024)." (p.5)
Q73	"The evaluations of LLMs use 5 different prompting templates and a temperature of 0.5." (p.6)

Q84	“Evaluation is based on accuracy and F1-score for ID and SF tasks. Average computed on five templates, and on only African languages.” (p.7)
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Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Yes — text output for intent + slots matches Telegram/SMS chatbot needs. No speech, image, or open-ended generation required.

OF-2: N/A — TTS not required; deployment is text-based.

OF-3: Adult literacy in Addis Ababa is high (male ~93.6%, female ~80% per 2005 baseline [WEB-2, WEB-4]) — text-based interaction is feasible for the target population. Telegram dominance and 4G coverage >80% [WEB-6] support text channel.

OF-4: Minor: `$$`-separated string output [Q146] requires production-side parsing logic; no probabilistic confidence scores in benchmark output schema, which is sub-optimal for dialogue-management uncertainty handling but does not invalidate evaluation.

ID	Evidence
Q84	“Evaluation is based on accuracy and F1-score for ID and SF tasks. Average computed on five templates, and on only African languages.” (p.7)
Q145	“Please ensure that the entities match the listed types and that unstated entities should not be included in the response if no entities are found, return `$$` only.” (p.19)
Q146	“Sentence: John went to Paris and paid 100 dollars at an Awater restaurant. Output: PERSONAL_NAME John $$ CITY_OR_PROVINCE Paris $$ MONEY 100 $$ RESTAURANT_NAME Awater” (p.19)
WEB-2	Addis Ababa adult literacy >93% male, ~80% female — text-based interaction feasible (https://worldpopulationreview.com/cities/ethiopia/addis-ababa)
WEB-4	Wikipedia Demographics of Addis Ababa corroborates literacy figures (https://en.wikipedia.org/wiki/Addis_Ababa)
WEB-6	4G coverage >80% in primary deployment cities — text channel infrastructure available (https://www.mordorintelligence.com/industry-reports/ethiopia-telecom-mno-market)

Information Gaps

- Whether F1 with macro vs micro averaging is reported by slot type or language
- Whether confidence scores can be extracted from fine-tuned models for production uncertainty handling